Shriiraz Kukatla

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.48
Narayana Junior college	Class XII (TSBIE)	2023	97.20%
Narayana Kukatpally school	Class X (BSE Telangana)	2021	10.00%

EXPERIENCE

Samsung -- Software Engineering Intern May 2026 - July 2026

Pulse | Telecommunications Diagnostic & Processing Platform

- Developed an automated **IQ data and sector-configuration extraction pipeline**, replacing a manual multi-tool workflow with a one-click backend process and reducing a ~2-hour data preparation workflow to an automated pipeline.
- Implemented **FFT-based guard-band removal** for LTE/5G IQ data while preserving .mbin file size and hardware data alignment.
- Built **binary-level algorithms for slot detection and Resource Block mapping**, deriving hardware data layouts from raw byte structures and telecom configuration parameters.
- Designed **asynchronous job orchestration and worker dependencies** for configuration parsing, IQ splitting, and signal-processing jobs.

SKILLS

- **Programming Languages:** C,C++,Python,Javascript, Assembly
- **Frameworks:** Express.js, Flask
- **Databases:** MySQL, MongoDB

PROJECTS

Infinite Precision Arithmetic

Course Project under Prof. Ramakrishna Upadrasta & Prof. Rajesh Kedia

- Designed and developed an **infinite-precision arithmetic library** in **C++** using OOP principles.
- Implemented custom Integer and Float classes supporting addition, subtraction, multiplication, and division with arbitrary precision.
- Ensured exact computation for large integers and floating-point numbers, avoiding round-off errors.

RISC-V Simulator

Course Project under Prof. Rajesh Kedia

- Developed a functional simulator for the **RISC-V instruction set** using **C++**.
- Implemented parsing and translation of assembly code into machine code instructions.
- Simulated instruction execution with complete register and memory modeling.

Operating Systems Memory Management

Course Project under Prof. Satya Peri & Prof. Saurabh Kumar

- Implemented custom **mmap**, **munmap**, and **mprotect** system calls in **gemOS** to manage process memory.
- Extended page table structures to handle virtual-to-physical address translation effectively.
- Added **Copy-on-Write** fork mechanism to optimize memory usage when creating new processes.

Qure -- Quantum Compiler

Course Project under Prof. Ramakrishna Upadrasta

- **Designed and implemented a high-level quantum programming language compiler in C++**, using Flex/Bison for lexical analysis and parsing and an **AST-based multi-pass compiler architecture**.
- **Implemented semantic analysis and type checking** for classical and quantum types, including scope management, qubit/bit validation, function and circuit constructs, and quantum-gate argument checking.
- **Built an AST optimization pipeline** with constant folding/propagation, dead-code elimination, compile-time loop unrolling, gate cancellation, and rotation-gate fusion, followed by **OpenQASM 3.0 code generation** for quantum execution.

Secure Multi-Tenant Agentic Workflow Sandbox

Course Project under Prof. Praveen Tammana

- Building an isolated execution platform for **untrusted Python and agent workloads**, with tenant isolation, resource limits, deny-by-default networking, and controlled tool/credential access.
- Designing a **stateful sandbox lifecycle** supporting admission control, checkpointing, cleanup, and restoration of paused sessions.
- Developing a **hybrid grading pipeline** combining deterministic tests with rubric-based LLM evaluation, confidence reporting, and human escalation.

RELEVANT COURSES

CS Courses: DS, Comp architecture, OS, DBMS, Theory of computation, Algorithms, Compilers, Computer networks, Machine learning, DL, NLP, Multi-Agent Systems, Distributed AI Training in a large-scale GPU cluster

Math courses: Probability and random variables, Linear algebra, Convex optimization, Computational Number Theory & Algebra

CERTIFICATIONS

- Completed Competitive Programming Summer Camp 2024, organized by **IICPC** (Inter IIT Competitive Programming Conclave).